Faculty of Engineering and Technology

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Computer Security - COMP432

Project Report

Facial Micro-Expressions

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Abstract

Research over the years has shown that humans are more alike than different, with genetic differences between individuals being only 3-5%. While this difference might seem small, it poses a significant challenge for cybersecurity professionals, who must develop reliable methods to distinguish individuals for secure authentication. Biometrics, which uses unique physical or behavioral traits, has become an essential tool in authentication systems. Common biometric methods include fingerprint scanning, iris recognition, and signature analysis. This project focuses on exploring facial micro-expressions—subtle, brief facial movements that reveal hidden emotions—as a new biometric method. In this report, we will examine the nature of facial micro-expressions, their unique characteristics, and their potential use in advanced authentication systems, while also discussing their benefits and challenges.

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Introduction

➢ Background on Micro-Expression Analysis

Emotions are the body's reactions to things happening around or inside us. They are associated with feelings, thoughts, behavioral responses, and pleasure or displeasure, which influence human thinking, decision-making, perception, learning, and so on. Therefore, emotions play an important role in everyday human life. However, emotional expression and understanding are not easy jobs for some people, such as those who suffer from psychological disorders, e.g., alexithymia. In recent years, research on emotions has grown significantly in interdisciplinary fields from psychology to computer science. Psychological research demonstrates the body language that we use, specifically our facial expressions, and relates to 55% of messages when people perceive others' feelings.

To this end, facial expressions are a major channel that humans use to express emotions. Analyzing facial expressions is meaningful and important, which can be seen from a wide study of facial expressions. However, people may try to conceal their true feelings under certain conditions when people want to avoid losses or gain benefits. In this case, facial micro-expressions (MEs) may occur.

● *What Are Micro-Expressions?*

Micro-expressions (MEs) are facial expressions that occur within a fraction of a second. This involuntary emotional leakage exposes a person’s true emotions. It occurs in high-stake situations when individuals try to hide or restrain their genuine emotions. Hence, MEs can provide essential clues to people’s true feelings

and have plenty of potential applications, such as national security, clinical diagnosis, and interrogations. In recent years, ME analysis has gained much attention in various fields due to its practical importance, especially automatic ME analysis in computer vision as MEs are difficult to process by naked eyes.

● *Historical Context: Discovery by Dr. Paul Ekman:*

The discovery of MEs was by Paul Ekman, who began his research on facial expressions and body movements, including hand gestures, in 1967. Dr. Ekman worked with clinical cases in which patients lied about their emotional state. He studied patients who claimed they were not depressed and later committed suicide. Upon examining films of the patients in slow motion, Dr. Ekman and Dr. Wallace Friesen spotted micro facial expressions, which revealed strong negative feelings the patient was trying to hide[1].

● *Micro-Expressions vs. Macro-Expressions:*

Micro-expressions (MEs) are different from macro-expressions (MaEs), which are the regular facial expressions we see every day. MaEs use several facial muscles, can have different levels of strength, and last between 0.5 and 4 seconds. In comparison, MEs use fewer muscles (usually just one or two), are much weaker, and are very quick, lasting less than 500 milliseconds. MaEs are easier to notice and can be faked, so they don\u2019t always show a person\u2019s true emotions. MEs, however, are automatic and show real feelings, but they are so fast and subtle that it is almost impossible to see them without special tools.

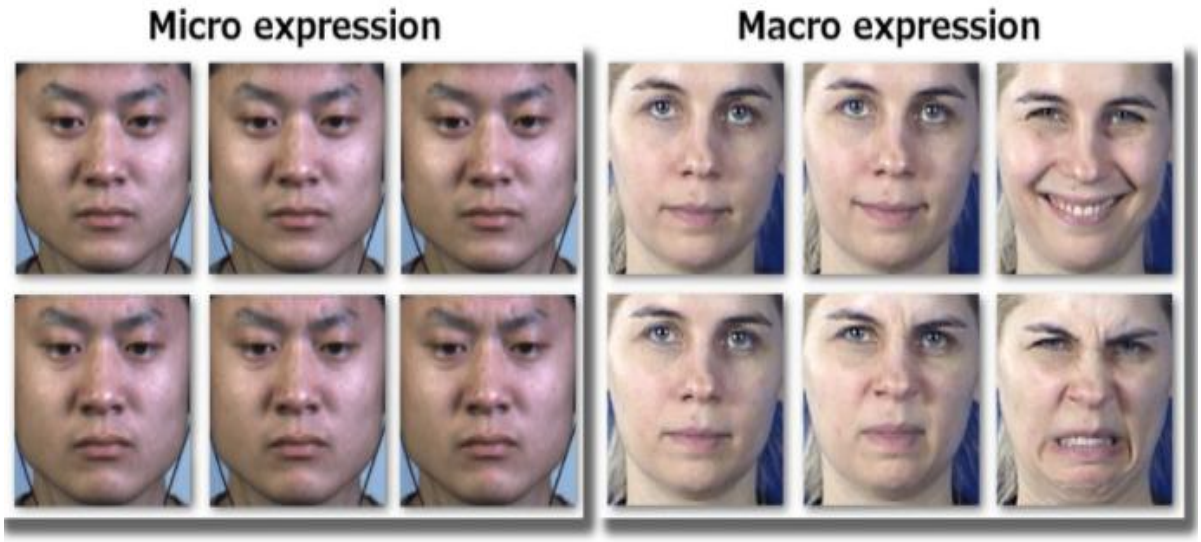

Figure1: Micro-Expressions vs. Macro-Expressions.

➢ Enhancing User Authentication with Micro Facial Expression Analysis:

The authors propose using micro expression analysis along with face recognition for user authentication. By combining micro expressions with face recognition, the accuracy of the authentication process can be improved, making it harder to fake someone's identity using photos or other deceptive techniques. For example, users could be shown images or text that trigger certain emotions, causing them to display micro expressions. Although this approach shows promise, there are challenges, such as users getting used to the emotional triggers or trying to fake their emotions. Therefore, methods need to be developed to avoid these issues[8].

➢ Importance in Security

The analysis and recognition of micro-expressions have become an essential part of various fields such as security and psychology. In security matters, micro-expressions are widely used to detect deception.

Due to the increasing need for deception detection in many areas such as criminal investigations, airport security, national security, interrogations, and interviews, the ability to detect deception effectively and reliably would have significant benefits for society.

Facial expressions can be used to detect deception through deception indicators by identifying the differences between involuntary and simulated expressions.

In the scientific literature, micro-expressions have been used to determine the mental status of patients with schizophrenia, to protect against thieves and malicious individuals, and to detect pain in newborn babies. Some studies use micro-expressions to detect driver fatigue and sleepiness, predict safe driving behavior, and create automatic music playlists[2].

Literature

➢ Overview of Facial Recognition in Security

In today's digital world, ensuring that only the right people have access to sensitive information is very important. Biometric facial recognition technology is leading the charge in this area, offering a simple yet secure way to verify identities and grant access. By using the unique features of people's faces, this technology not only makes things safer but also smoother across different industries.

With biometric facial recognition, there's no need to remember complex passwords or PINs. Instead, face recognition technology scans your face to confirm who you are, making the process both convenient and secure. As organizations focus more on protecting data and respecting privacy, biometric facial recognition is becoming a trusted tool for verifying identities and controlling access.

Biometric facial recognition significantly enhances user experience by creating a seamless and efficient means of access control. In a time when worries about fraud, data security, and privacy are rising, biometric facial recognition presents itself as a reliable method for strengthening the protection of confidential data and guaranteeing that only those with approved access rights are allowed access. With its integration of advanced biometric identification techniques and sophisticated facial recognition algorithms, facial recognition software serves as a powerful and trusted guardian of digital identities, safeguarding against potential fraud and violations.

Face recognition is the most natural and widely studied method for identifying people, but it struggles in uncontrolled environments. Researchers have tried combining appearance-based traits like facial marks and hair color with behavior-based traits from videos, such as head movements and facial expressions. Initially seen as a source of variation, facial expressions are now being used to enhance recognition as a soft biometric.

However, facial expressions can be easy to fake, so they may not be reliable for identifying people. Micro-expressions, on the other hand, are almost impossible to replicate and could be useful for recognizing individuals and preventing spoofing attacks. They also offer better privacy protection because they aren't detailed

enough to reveal personal traits like age or gender. This makes them a safer option for biometric systems[3].

➢ Key Studies on Micro-Expressions & Emotion Detection

Facial micro-expressions are universal emotional cues that enhance communication and play a key role in emotion psychology. They are important in areas like communication, psychology, human-computer interaction, medicine, virtual reality, and intelligent systems.

This study aims to detect deception by analyzing video modalities that are collected from human faces such as facial expressions. To test the custom-trained deception detection model's performance, several experiments were conducted and it achieved 74.17% correct classification accuracy of deception.

Humans struggle to recognize micro-expressions accurately, making it necessary to develop automated systems. These systems use computer-based methods to detect both macro- and micro-expressions, which can show seven emotions—sadness, happiness, fear, anger, surprise, disgust, and contempt.

Machine learning algorithms have been used in previous studies for various classification problems, however, only a few studies have been carried out in this area. In the existing studies, supervised machine learning algorithms were used which include decision trees, neural networks, k nearest neighbors (KNN), Support Vector Machine (SVM), ELM and random forest classifiers. Support Vector Machine appears to be the most used machine learning model for facial expression recognition due to its strong ability to generalize well, even with imbalanced training samples[4].

Milestones

➢ Timeline and Progress Overview

The process of micro-expression recognition can be divided into three main phases: data collection/preparation, feature extraction, and classification.

Micro-expression datasets are categorized as either acted or spontaneous. Acted samples are performed by subjects following instructions, while spontaneous samples capture genuine emotions triggered in real time, making them more authentic. Examples of acted datasets include the USFHD and Polikovsky's databases, while spontaneous datasets include SMIC, CASME, and CASME II. This study uses CASME II, as it is the most recent, publicly available, and widely used for research.

The CASME II database is an improved version of the CASME dataset, containing 247 micro-expression samples from 26 participants. Emotional responses were recorded using a high-resolution camera as participants watched emotional video clips and rated their intensity on a 7-point scale. The recordings were divided into frames, pre-processed to remove irregular movements, and selected based on durations under 500 milliseconds. Samples were labeled using the Facial Action Coding System (FACS) and categorized into five classes: disgust, happiness, surprise, repression, and others.

Before release, the CASME II database underwent preprocessing, including face detection, video frame division, and labeling with action units. It also marks the onset, apex, and offset frames—onset being the first sign of change, apex showing

peak intensity, and offset marking the return to neutral. The database contains 247 samples recorded at 200 fps.

Micro-expression (ME) analysis requires preprocessing to address issues such as background removal, head pose changes, and variations in face size and shape. Since MEs are subtle, failing to handle these factors can impact performance. Common preprocessing steps include face detection and alignment.

The machine learning technique explored in this study is Extreme Learning Machine (ELM) due to its fast learning speed and high performance. Support Vector Machine (SVM) is used as a baseline for comparison. Features are extracted using Local Binary Pattern (LBP) for apex frames and LBP-TOP for motion-based features. Evaluations on the CASME II dataset show that ELM outperforms SVM in accuracy, precision, recall, and F-score, especially with temporal features. ELM also trains faster, averaging 0.0409 seconds compared to SVM's 0.3405 seconds. The results highlight the effectiveness of fast-learning algorithms and temporal features in micro-expression recognition.

Current micro-expression recognition (MER) research faces challenges due to the reliance on lab-controlled datasets, which lack the complexity of real-world scenarios. Real-world conditions involve factors such as lighting variations, 3D head movements, and multi-person interactions, making MER more difficult. Lai et al. developed a real-time MER framework running at 60 fps on high-performance hardware, but it was tested only on lab datasets (CASME and CASME II), leaving MER in the wild as an open challenge.

To improve performance, future research should address real-world conditions, optimize computational efficiency, and ensure privacy compliance. Ethical concerns regarding data collection, such as informed consent and emotional stimulation, must also be prioritized, as there are currently no unified guidelines to handle these issues[5].

Discussion

➢ Analysis of Findings

The interpretation of results for micro-expressions in computer security focuses on how these expressions enhance security systems, and demonstrate the potential benefits of this technology.

1. Attacker Intent Detection: Analyze facial expressions such as anxiety to detect hacking or fraud attempts during identity verification processes.

2. Better Security Systems: Using micro-expressions with biometric authentication makes security systems more accurate and helps detect tampering more reliably.

3. Understanding Behavior: Watching small emotional changes helps to better understand a person's actions, which is important for finding threats in places like airports or banks.

4. Quick Threat Response: Systems that can quickly study facial expressions can act faster to stop unusual or dangerous behavior.

By analyzing these facial expressions, security systems can become smarter and more responsive, helping to spot threats or fraud before they happen.

➢ Challenges and Limitations

The use of facial micro-expressions in security faces many limitations and challenges that affect its effectiveness, requiring continuous improvements to ensure accurate real-time analysis and system adaptability to meet growing security needs.

1. Detection Difficulty:

Micro-expressions are brief and subtle, making them hard to detect and analyze even with advanced technology.

2. Cultural differences:

Mean that people from different cultures may show the same feelings using different facial expressions or may interpret certain expressions in different ways. For example, one person from a certain culture might think a specific expression shows anger, while someone from another culture might see it as surprise or sadness. So, systems need to be developed in a way that considers these differences.

3. Technical Constraints:

Detecting micro-expressions requires accurate algorithms, as these expressions appear in a small area of the face and last for a very short time. This makes them challenging to capture and analyze accurately, demanding highly precise and powerful techniques.

4. Data Limitations:

Means there aren't enough diverse examples of images or videos showing micro-expressions in different situations. To train a system to recognize these expressions accurately, it needs many examples, including people from different cultures, ages, and environments (like different lighting or camera angles). Without enough varied data, it's hard to train systems to recognize expressions correctly in real-life situations.

5. Privacy Concerns:

Involve ethical issues related to using micro-expression analysis without consent, especially in areas like security or surveillance. This could mean monitoring individuals without their knowledge or making decisions based on their emotions without their approval.

6. Environmental Factors:

Refer to external conditions that can impact the ability to accurately detect micro-expressions. For example:

- Lighting: Poor or uneven lighting can make it difficult to see facial expressions.
- Facial Obstructions: Things like hair, glasses, or anything covering parts of the face can block or hide expressions.
- Camera Angles: If the camera is at the wrong angle or too far away, it might miss subtle facial movements or distort them.

All of these factors affect the accuracy of detecting micro-expressions[6].

Further Work

➢ Future Research Directions

Facial micro-expression analysis has the potential to greatly improve security systems. Future research should focus on enhancing accuracy, efficiency, and adaptability in situations.

1. Building Better Algorithms:

Work on creating smarter algorithms that can detect micro-expressions more accurately, even with challenges like cultural differences, bad lighting, or obstacles like glasses and masks.

2. Integration with Multimodal Systems:

Create systems that use micro-expressions along with other information, like voice tone, body movements, and eye behavior, to get a fuller understanding of a person's emotions or state.

3. Adapting to Dynamic Environments:

Improve systems to work well in crowded places like airports or train stations and with people who are moving quickly.

4. Making Ethical Rules:

Set rules to ensure these technologies are used responsibly, protecting privacy and individual rights, and preventing misuse in ways that harm or monitor people without their approval.

5. Real-Time Processing:

Develop faster and more efficient methods to analyze facial expressions instantly, enabling systems to respond quickly in real-world situations.

➢ Enhancing Real-World Applications

Micro-expressions are a powerful tool, that enables improved security and interaction with smart systems in ways that were not possible before.

1. Security Surveillance: Using micro-expression analysis in smart cameras to detect suspicious behavior in sensitive places like airports and government facilities.

2. Fraud Prevention: Applying the technology in banks and online transactions to detect fraudulent activities by analyzing emotional signals.

3. Mental Health Care: Helping doctors understand patients' emotional states better to provide effective treatments.

4. Smart System Interaction: Improving user experience with voice assistants and robots by responding to emotional cues.

5. Workplace Monitoring: Detecting stress or dissatisfaction among employees to boost productivity and well-being.

6. Smart Vehicles and Driving: Monitoring drivers' states to detect fatigue or stress, ensuring safer driving.

These applications are widely used and have a big impact, as they improve important areas like security, health, and technology interaction, showing great potential to change many fields[7].

Conclusion:

In conclusion, facial micro-expressions emerge as an innovative solution that could redefine how we verify identities. These small but powerful emotional signals may hold the key to protecting sensitive information and preventing increasingly complex digital attacks. While there are challenges in using them in real-world environments, the opportunities they offer in enhancing security and providing reliable, secure authentication methods cannot be ignored. As research and technology continue to advance, facial micro-expressions could become an essential part of future biometric systems, opening new doors to a more interactive and accurate security experience. Therefore, this technology could become the cornerstone of identity verification solutions shortly.

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